

Optimal Non-Adaptive Group Testing under Markovian Infection Models

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Abstract—THIS PAPER IS ELIGIBLE FOR THE STUDENT PAPER AWARD. We consider group testing under a Markov infection model, in the sparse-infection regime where the expected number of infected items scales as $O(n^\theta)$ for some $\theta \in (0, 1)$. In this setting, we exactly characterize the optimal number of non-adaptive tests. The resulting result generalizes the independent-prior based results of Coja-Oghlan et al. [1] to a nontrivial setting with correlated priors. Our results imply that Markovian correlations reduce the non-adaptive testing capacity in sufficiently sparse regimes. However, they also identify a threshold in the sparsity parameter θ above which correlations provide no improvement in the non-adaptive testing capacity.

I. INTRODUCTION

We consider a *correlated* group testing problem in which the infection vector is generated by a Markov source. We investigate the non-adaptive [2] testing capacity of this model.

Group testing was introduced in [3] as a strategy for reducing the number of tests needed to estimate disease prevalence in large populations, when infections are sparse. A classical formulation is the independent combinatorial model, in which a population of size n contains exactly k infected individuals, chosen uniformly at random from all subsets of size k . Formally, the infection vector is denoted by $U^n \in \{0, 1\}^n$, where $U_i = 1$ indicates that item i is infected. A non-adaptive test design is represented by a binary matrix $X \in \{0, 1\}^{T \times n}$, where all tests are fixed before observing any outcomes, and $X_{t,i} = 1$ indicates that item i is included in test t . The corresponding noiseless test outcomes are given by

$$Y_t = \mathbf{1} \left\{ \sum_{i=1}^n X_{t,i} U_i > 0 \right\}, \quad t = 1, \dots, T.$$

Thus, a test returns positive if it contains at least one infected item. The goal is to construct an estimator $\hat{U}^n = \hat{U}^n(X, Y)$ that recovers U^n from the test design and outcome vector.

This naturally led to interest in characterizing the fundamental limits on the number of tests required for reliable recovery of the infected individuals. The optimal number tests under such independent combinatorial infection models has since been studied extensively [2], [4]–[8]. In particular, the non-adaptive capacity under the independent combinatorial model was resolved by Coja-Oghlan et al. in [1], [9].

A natural motivation for the correlated setting is that independence rarely reflects real-world infection processes. Infection vectors are often better modeled by correlated sources,

where an item’s infection status depends on the infection status of other items, rather than being independent across items. For example, individuals in the same household, neighborhood, or community are more likely to be infected together, producing clustered or bursty infection patterns [10]–[12]. Similar correlation structures also arise in biological testing applications, such as spatial transcriptomics [13], where co-regulated genes may be activated together. These settings motivate a theory of non-adaptive group testing under correlated infection priors.

Many recent works have sought to relax the independence assumption. One line of work studies community-based infection models, in which infections spread over predefined graph structures [14]–[18]. Other works consider alternative correlated priors, including Markovian infection models [19], [20], non-uniform priors [21] and hypergraph models [22].

The optimal number of tests in the non-adaptive setting remains poorly understood for most correlated models. For example, [14], [15] characterize rates under a constant, non-vanishing false-positive rate, but do not provide a sharp threshold. Similarly, [19] focuses on computationally efficient decoders whose testing rates remain strictly suboptimal.

As a concrete instance of a correlated setting with this technical obstacle, we consider the Markovian infection model introduced in [19], considered in Figure 1. In this model, the infection vector U^n is generated by a two-state Markov chain, initialized according to its stationary distribution. The transition probabilities are α_n and β , where $\alpha_n = k'n^{\theta-1}$ for some constant k' and $\beta \in (0, 1)$ is a constant. Thus, infections occur in bursts: once the chain enters state 1, it remains infected for a geometrically distributed number of consecutive positions with mean $1/\beta$. The stationary marginal infection probability scales as $q_n \asymp n^{\theta-1}$, and hence the expected number of infected items scales as $nq_n \asymp n^\theta$, consistent with the sparse regime we consider in this paper.

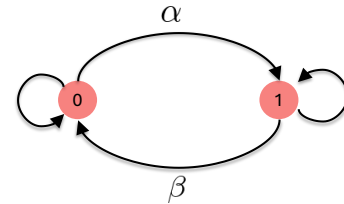


Fig. 1. The 2-state Markov chain that generates the infection vector U^n

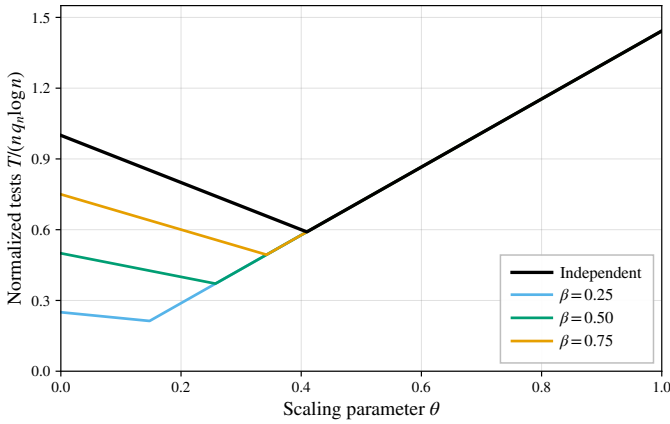


Fig. 2. Normalized number of tests required for exact recovery under the independent prior and correlated Markov priors with different values of β . Correlations reduce the testing threshold in sparse regimes, while all curves coincide with the independent threshold $\theta/\ln 2$ once θ is sufficiently large.

Under this model, we establish a sharp testing threshold T_{cap} , where

$$T_{\text{cap}} = \max \left\{ \beta(1 - \theta), \frac{\theta}{\ln 2} \right\} nq \log n,$$

where $\log$ is base 2. Reliable recovery of U^n is asymptotically possible if T is above this threshold and impossible below it.

Figure 2 compares the normalized number of tests required under the independent and Markovian infection models. In particular, when $\theta > \ln 2 / (1 + \ln 2)$, the Markovian correlation does not improve the testing capacity relative to the independent case. As $\beta \rightarrow 1$, the Markovian model approaches the independent-prior setting, and the resulting threshold recovers the corresponding independent group testing threshold.

The main technical contribution is a new source-dependent typical set, which captures the high-probability infection vectors generated by the prior. Unlike in the classical combinatorial model, competitors at a fixed overlap cannot be counted by binomial arguments. We develop sharp large-deviation and Sanov-type counting bounds for these typical sequences and combine them with test-design collision probabilities. This yields a decoupling principle: the error probability separates into a source-complexity term, controlling the number of plausible competitors, and a test-design term, controlling the probability that such competitors match the observed outcomes. This decoupling enables the exact characterization of the optimal non-adaptive testing threshold.

II. PROBLEM FORMULATION AND PRELIMINARIES

We consider the Markovian correlated group testing problem studied in [19], in which n distinct items, indexed by $\{1, 2, \dots, n\}$, are assigned binary labels $U_i \in \{0, 1\}$.

The two-state Markov chain has a state space $S = \{0, 1\}$ and transition probabilities $p_{0,1} = \alpha$ and $p_{1,0} = \beta$, as shown in

Figure 1. The infection vector U^n is generated by initializing the chain in its stationary distribution

$$(q, 1 - q) := \left(\frac{\alpha}{\alpha + \beta}, \frac{\beta}{\alpha + \beta} \right), \quad (1)$$

where $q := q_n$. Then we record n consecutive states.

Let $T := T(n)$ denote the number of tests. A non-adaptive test design is specified by a binary matrix $X \in \{0, 1\}^{T \times n}$, chosen before any test outcomes are observed, where $X_{t,i} = 1$ indicates that item i is included in test t . The noiseless outcome vector $Y = (Y_1, \dots, Y_T)$ is then given by

$$Y_t = \mathbf{1} \left\{ \sum_{i=1}^n X_{t,i} U_i > 0 \right\}, \quad t = 1, \dots, T. \quad (2)$$

Equivalently, $Y_t = 1$ if t contains at least one infected item. A decoding rule G then maps (X, Y) to an estimate $\hat{u}^n \in \{0, 1\}^n$. Let $\mathcal{E} = \{U^n \neq \hat{U}^n\}$ denote the error event.

The objective is to construct a sequence of non-adaptive test matrices and associated decoders such that $\Pr(\mathcal{E}) \rightarrow 0$ as $n \rightarrow \infty$. Following [4], we study the regime in which the number of tests scales on the order of $nq_n \log n$. Based on this scaling, we define an achievable number of normalized tests as follows:

Definition 1 (Achievability): A normalized number of tests τ is said to be *achievable* if there exists a sequence of non-adaptive test matrices $X_n \in \{0, 1\}^{T(n) \times n}$ and decoding rules $g_n : \{0, 1\}^{T(n) \times n} \times \{0, 1\}^{T(n)} \rightarrow \{0, 1\}^n$ such that, with $\hat{U}^n = g_n(X_n, Y)$,

$$\lim_{n \rightarrow \infty} \frac{T(n)}{nq_n \log n} = \tau, \quad (3)$$

and $\Pr(\mathcal{E}) \rightarrow 0$ as $n \rightarrow \infty$. The optimal normalized number of tests is defined as

$$\tau^* := \inf \{ \tau : \tau \text{ is achievable} \}. \quad (4)$$

The infimum is taken over all sequences of non-adaptive test designs $\{X^{(n)}\}_{n \geq 1}$ and all corresponding decoders $\{g_n\}_{n \geq 1}$. Equivalently, for any $\tau < \tau^*$, there is no sequence of test designs and decoders for which the error probability vanishes as $n \rightarrow \infty$. We note that while this definition is stated differently from the capacity formulation in [4], the two definitions characterize the same fundamental threshold.

Following the group testing literature, we work in the sparse regime [4], where the expected number of infected items satisfies $nq_n = \Theta(n^\theta)$, $\theta \in (0, 1)$. We specify

$$\alpha = \frac{k' n^\theta}{n}, \quad (5)$$

with $k' > 0$ constant and $\beta \in (0, 1)$ fixed. Then

$$q_n = \frac{\alpha}{\alpha + \beta} = \frac{k'}{\beta} n^{\theta-1} + o(n^{\theta-1}). \quad (6)$$

Writing $k := k'/\beta$, we use the shorthand $q_n = kn^\theta/n$, omitting lower order terms, since they do not affect results.

We introduce a notion of typical infection vectors tailored to the sparse Markov source. This construction is based on a re-normalized entropy rate, which captures the source complexity.

Lemma 1 (Sparse entropy rate): For the Markov infection source described above,

$$\frac{H(U^n)}{nq_n \log n} = \beta(1 - \theta) + o(1).$$

We then define a typical set in terms of this entropy rate.

Definition 2 (ϵ -typical set): For $\epsilon > 0$, define

$$T_\epsilon^{(n)} := \left\{ u^n : \left| -\frac{1}{nq_n \log n} \log P_{U^n}(u^n) - \beta(1 - \theta) \right| \leq \epsilon, \right. \\ \left. \left| \frac{1}{n} \sum_{i=1}^n u_i - q_n \right| \leq \epsilon \right\}. \quad (7)$$

The first condition states that every sequence in $T_\epsilon^{(n)}$ has probability close to the typical probability scale of the source. The second condition enforces the sparse weight constraint: every typical sequence has approximately nq_n infected entries.

Lemma 2 (Typical-set concentration): Consider the Markov infection source described above, and let $T_\epsilon^{(n)}$ be the ϵ -typical set in Definition 2. Then, for every fixed $\epsilon > 0$,

$$\mathbb{P}(U^n \in T_\epsilon^{(n)}) \rightarrow 1 \quad \text{as } n \rightarrow \infty.$$

The proofs of Lemmas 1 and 2 are provided in a longer version of this paper.

Notation. Throughout, $\log$ denotes base-2 logarithm and $\ln$ denotes the natural logarithm. We use the standard asymptotic notation $o(\cdot)$ and $\Theta(\cdot)$ as $n \rightarrow \infty$. Throughout, for any positive integer m , we use the notation $[m] := \{1, 2, \dots, m\}$.

III. TEST DESIGN AND MAIN RESULTS

Before stating the full result, we describe the test matrix design and decoding rule used throughout the achievability analysis. The construction largely follows the near-constant-column design and consistency decoder considered in [9].

Recall that $X \in \{0, 1\}^{T \times n}$ denotes the non-adaptive test matrix. Fix a design parameter $\nu > 0$ and set

$$\Delta := \frac{\nu T}{nq_n}.$$

For each item $i \in [n]$, independently choose Δ test indices uniformly at random from $[T]$, with replacement, and set $X_{t,i} = 1$ if item i is assigned to test t at least once.

For any $u^n \in \{0, 1\}^n$, let $Y(u^n)$ denote the noiseless outcome vector induced by u^n under X . Given X and Y , the decoder searches for vectors in the typical set that are consistent with the observed outcomes:

$$\mathcal{C}(Y) := \left\{ u^n \in T_\epsilon^{(n)} : Y(u^n) = Y \right\}.$$

If $\mathcal{C}(Y) = \{u^n\}$ is a singleton, the decoder returns $\hat{U}^n = u^n$. Otherwise the decoder declares an error.

We now state our main theorem which characterizes the optimal normalized number of tests.

Theorem 3 (Optimal normalized number of tests): Consider the sparse regime $nq_n = n^\theta$, where $\theta \in (0, 1)$, and let τ^* be the optimal normalized number of tests. Then

$$\tau^* = \max \left\{ \beta(1 - \theta), \frac{\theta}{\ln 2} \right\}.$$

The threshold has two distinct contributions. The term $\beta(1 - \theta)$ is the source-complexity term, which reflects the number of tests needed to distinguish the true infection vector from the collection of other typical infection vectors. The term $\theta / \ln 2$ is a high-overlap indistinguishability term, which arises from competitors that differ from the true vector in only a small number of infected coordinates and are therefore difficult to separate through pooled tests. A related obstruction was identified by [9] in the independent prior setting. The key technical challenge for this term is to show that the same obstruction persists under the Markov source, despite the correlations between infection coordinates and the non-uniform structure of the source. We prove the upper bound in Section IV and the lower bound in Section V.

IV. ACHIEVABILITY

We now outline the achievability argument under the constant test design and the typicality-constrained decoder. Define the overall error event $\mathcal{E} := \mathcal{E}_0 \cup \mathcal{E}_1$, where

$$\mathcal{E}_0 = \left\{ U^n \notin T_\epsilon^{(n)} \right\}, \\ \mathcal{E}_1 = \left\{ \exists v^n \in T_\epsilon^{(n)}, v^n \neq U^n : Y(v^n) = Y(U^n) \right\}. \quad (8)$$

By the axiom of probability,

$$\mathbb{P}(\mathcal{E}) = \mathbb{P}(\mathcal{E}_0) + \mathbb{P}(\mathcal{E}_1 | \mathcal{E}_0^c). \quad (9)$$

Since $T_\epsilon^{(n)}$ is a typical set satisfying Lemma 2, for all sufficiently large n , we have

$$\mathbb{P}(\mathcal{E}) \leq \mathbb{P}(\mathcal{E}_1 | U^n \in T_\epsilon^{(n)}) + \epsilon. \quad (10)$$

We now bound this by grouping competitors v^n according to their “infected-overlap” with the realized sequence U^n . For two fixed sequences $u^n, v^n \in \{0, 1\}^n$, let

$$l(u^n, v^n) := \sum_{i=1}^n \mathbf{1}\{u_i = 1, v_i = 1\} \quad (11)$$

denote the number of coordinates at which both sequences are equal to 1. Recall $k = nq$, and define $\lambda(u^n, v^n) := \frac{l(u^n, v^n)}{k + \epsilon}$, where λ is the normalized overlap. For each value of λ , let

$$N(\lambda | u^n) := \left| \left\{ v^n \in T_\epsilon^{(n)} \setminus \{u^n\} : \frac{l(u^n, v^n)}{k + \epsilon} = \lambda \right\} \right|,$$

denote the number of sequences $v^n \in T_\epsilon^{(n)}$ with $v^n \neq u^n$ and normalized overlap λ with u^n . Let $\lambda_n^* := 1 - \frac{1}{\log n}$ be the overlap threshold used to separate the small-overlap and large-overlap regimes. We now decompose the error event according to the normalized overlap between the true sequence u^n and a competing sequence v^n . Let $p_T(u^n) := \mathbb{P}(U^n = u^n | U^n \in T_\epsilon^{(n)})$. For overlap levels $\lambda \leq \lambda_n^*$, we union bound

over all competitors in that overlap class. The number of such competitors is denoted by $N(\lambda \mid u^n)$. All competitors with overlap larger than λ_n^* are collected into a single large-overlap event. Thus,

$$\begin{aligned} \mathbb{P}(\mathcal{E}_1 \mid U^n \in T_\epsilon^{(n)}) &= \sum_{u^n \in T_\epsilon^{(n)}} p_T(u^n) \mathbb{P}(\mathcal{E}_1 \mid U^n = u^n) \\ &\leq \sum_{u^n \in T_\epsilon^{(n)}} p_T(u^n) \left[\sum_{\lambda \leq \lambda_n^*} N(\lambda \mid u^n) \mathbb{P}(\mathcal{C}(u^n, v^n) \mid \mathcal{O}_\lambda(u^n, v^n)) \right. \\ &\quad \left. + \mathbb{P}(\mathcal{C}_{>\lambda_n^*}(u^n) \mid U^n = u^n) \right]. \end{aligned} \quad (12)$$

Here, in the first term, v^n denotes an arbitrary typical competitor in the overlap class

$$\mathcal{O}_\lambda(u^n, v^n) := \{u^n, v^n \in T_\epsilon^{(n)} : \lambda(u^n, v^n) = \lambda\},$$

and

$$\mathcal{C}(u^n, v^n) := \{Y(v^n) = Y(u^n)\}$$

is the event that this fixed competitor produces the same test outcome as u^n . The large-overlap event is

$$\mathcal{C}_{>\lambda_n^*}(u^n) := \left\{ \exists v^n \in T_\epsilon^{(n)} \setminus \{u^n\} : \begin{array}{l} \lambda(u^n, v^n) > \lambda_n^*, \\ Y(v^n) = Y(u^n) \end{array} \right\}.$$

We control the first term in (12) by combining two bounds. First, we bound $N(\lambda \mid u^n)$, the number of typical competitors with overlap λ . Second, we bound the probability that one fixed competitor in this overlap class produces the same test outcome as u^n . We control the second term directly by showing that the near-constant-column design rules out all typical competitors with overlap larger than λ_n^* . We now state the corresponding bounds and give proof sketches.

Lemma 3: [Expected $N(\lambda)$ bound] Fix $\epsilon > 0$. Then

$$\mathbb{E} \left[N(\lambda \mid U^n) \mid U^n \in T_\epsilon^{(n)} \right] \leq 2^{nq \log n [\beta(1-\theta)(1-\lambda) + \epsilon + o_n(1)]}.$$

Proof Sketch: Fix $u^n \in T_\epsilon^{(n)}$ and condition on an overlap level λ . A competing sequence v^n with $\lambda(u^n, v^n) = \lambda$ must agree with u^n on roughly λnq infected coordinates and introduce roughly $(1-\lambda)nq$ new infected coordinates outside the support of u^n .

Under the Markov source, infected coordinates occur in runs. The number of infected runs in a typical sequence is approximately βnq , because each infected run ends through a $1 \rightarrow 0$ transition, which occurs with probability β . Thus, replacing a $(1-\lambda)nq$ new infections corresponds, up to lower-order effects, to replacing $\beta(1-\lambda)nq$ infected runs.

Each new infected run must begin with a $0 \rightarrow 1$ transition. Since such a transition occurs with probability $\alpha_n = \Theta(n^{\theta-1})$, the number of admissible places in the competitor for the start of new infected runs is $\approx \frac{1}{\alpha_n} = \Theta(n^{1-\theta})$.

Since there are at most $\beta(1-\lambda)nq$ new infected runs, the number of typical competitors with overlap λ is bounded by

$$N(\lambda \mid u^n) \leq (n^{1-\theta})^{\beta(1-\lambda)nq}$$

up to subexponential factors. Equivalently,

$$N(\lambda \mid u^n) \leq 2^{nq \log n [\beta(1-\theta)(1-\lambda) + \epsilon + o_n(1)]}.$$

The full proof is in the longer version of this paper [23].

Theorem 4: [Small-overlap probability] Fix $u^n, v^n \in T_\epsilon^{(n)}$ with $\lambda(u^n, v^n) = \lambda \leq \lambda_n^*$. Under the above test design,

$$\mathbb{P}(Y(v^n) = Y(u^n) \mid \mathcal{O}_\lambda(u^n, v^n)) \leq 2^{-T(1-\lambda)[1-\epsilon-o_n(1)]}.$$

Proof Sketch: Here we give a sketch for an i.i.d. Bernoulli test design, where each item is included in each test independently with probability $p = \nu/k$. The near-constant-column design creates mild dependencies between rows. These dependencies are handled in the full proof and do not change the final result.

Fix $u^n, v^n \in T_\epsilon^{(n)}$ with normalized overlap λ . For clarity, ignore the vanishing typicality deviations and assume $|\text{supp}(u^n)| = |\text{supp}(v^n)| = k$, $|\text{supp}(u^n) \cap \text{supp}(v^n)| = \lambda k$. Thus, $|\text{supp}(u^n) \cup \text{supp}(v^n)| = (2-\lambda)k$. A test separates u^n and v^n if it is positive for exactly one of the two vectors. This can happen in two ways. First, the test may be positive under u^n and negative under v^n . This means that the test avoids all infected items in v^n , but hits at least one infected item in $u^n \setminus v^n$. The probability of this event is

$$(1-p)^k - (1-p)^{(2-\lambda)k}.$$

Here, $(1-p)^k$ is the probability of avoiding all infected items in v^n , and $(1-p)^{(2-\lambda)k}$ is the probability of avoiding all infected items in $\text{supp}(u^n) \cup \text{supp}(v^n)$. Subtracting removes the case where the test also misses all infected items in $u^n \setminus v^n$. By symmetry, the probability that the test is positive under v^n and negative under u^n is also $(1-p)^k - (1-p)^{(2-\lambda)k}$. These two separating events are disjoint. Therefore, the probability that one test separates u^n and v^n is $2(1-p)^k - 2(1-p)^{(2-\lambda)k}$. Consequently, the probability that one test gives the same outcome under u^n and v^n is $1 - 2(1-p)^k + 2(1-p)^{(2-\lambda)k}$. Now choose $p = \ln(2)/k$. When $\lambda \leq \lambda_n^*$ we can say (under asymptotic approximations) that $(1-p)^k \approx \frac{1}{2}$ and $(1-p)^{(2-\lambda)k} \approx 2^{-(2-\lambda)}$. Hence the one-test consistency probability satisfies $1 - 2(1-p)^k + 2(1-p)^{(2-\lambda)k} = 2^{-(1-\lambda)}$. Since tests are independent under the Bernoulli design, the probability that all T tests give the same outcome under u^n and v^n is at most $(2^{-(1-\lambda)})^T$. Therefore,

$$\mathbb{P}(Y(v^n) = Y(u^n) \mid \mathcal{O}_\lambda(u^n, v^n)) \leq 2^{-T(1-\lambda)[1-o_\epsilon(1)-o_n(1)]}.$$

A full technical proof is deferred to the longer version [23].

Combining Theorems 3 and 4 gives the following small-overlap contribution.

$$\begin{aligned} &\sum_{u^n \in T_\epsilon^{(n)}} p_T(u^n) \sum_{\lambda \leq \lambda_n^*} N(\lambda \mid u^n) \mathbb{P}(Y(v^n) = Y(u^n) \mid \mathcal{O}_\lambda) \\ &\stackrel{(a)}{\leq} \sum_{u^n \in T_\epsilon^{(n)}} p_T(u^n) \sum_{\lambda \leq \lambda_n^*} 2^{nq \log n [\beta(1-\theta)(1-\lambda)]} 2^{-T(1-\lambda)} \\ &\stackrel{(b)}{\leq} \sum_{\lambda \leq \lambda_n^*} 2^{-(1-\lambda)[T-\beta(1-\theta)nq \log n][1-\epsilon-o_n(1)]} \\ &\leq (nq+1) 2^{-(1-\lambda_n^*)[T-\beta(1-\theta)nq \log n][1-o_\epsilon-o_n(1)]} \rightarrow 0, \end{aligned}$$

as $n \rightarrow \infty$, provided $T > \beta(1 - \theta)nq \log n$. Here, (a) uses the Lemmas 3 and 4 and (b) is due to the fact that $\sum_{u^n \in T_\epsilon^{(n)}} p_T(u^n) \leq 1$.

It remains to control the large-overlap term. For this part, we directly use the argument from [9] for the near-constant-column design.

Theorem 5: [Large-overlap error bound] Let $\lambda_n^* = 1 - 1/\log n$ and choose $\nu = \ln 2$. For every $u^n \in T_\epsilon^{(n)}$,

$$\mathbb{P}(\mathcal{C}_{>\lambda_n^*}(u^n) \mid U^n = u^n) \leq n^{\theta - \frac{T \ln 2}{nq \log n} + o(1)}.$$

Consequently, if $T > (1 + \delta) \frac{\theta}{\ln 2} nq \log n$ for some fixed $\delta > 0$, then

$$\sum_{u^n \in T_\epsilon^{(n)}} p_T(u^n) \mathbb{P}(\mathcal{C}_{>\lambda_n^*}(u^n) \mid U^n = u^n) \rightarrow 0.$$

Above the threshold, each infected item is isolated in many tests with high probability. Hence, any competitor that removes such an item must add enough infected items to keep those tests positive. A large-overlap competitor changes too few infected coordinates to do this, so any consistent competitor must have overlap at most λ_n^* . Thus, the large-overlap error is ruled out by the argument of [1]. We revisit this in a longer version of the paper [23].

Combining the small-overlap and large-overlap bounds shows that $\tau > \tau^*$ is achievable. This proves the achievability part of Theorem 3.

V. CONVERSE

We now prove the converse part of Theorem 3. The lower bound has two components. The first comes from the counting argument revisited in [19], and the second follows [1].

The lower bound corresponding to the first argument in the maximum in (13) follows from the source-coding bound of [24]. The bound gives

$$\tau^* \geq \lim_{n \rightarrow \infty} \frac{H(U^n)}{nq \log n}, \quad (13)$$

where $H(U^n)$ is the entropy of the infection vector. This entropy can be computed as [19], yielding $\tau^* \geq \beta(1 - \theta)$.

It remains to obtain the second term. Fix an arbitrary non-adaptive test design $X \in \{0, 1\}^{T \times n}$ and a true infection vector u^n . An infected item i is called *disguised* if every test containing i also contains some other infected item. For $b \in \{0, 1\}$, define the set of disguised items with status b by

$$V_b^+(X, u^n) := \left\{ i : \begin{array}{l} u_i = b, \forall t \in [T] \text{ with } X_{ti} = 1, \\ \exists j \neq i \text{ such that } X_{tj} u_j = 1 \end{array} \right\}.$$

Thus, every test containing an item in $V_b^+(X, u^n)$ is already positive because of some infected item other than that item itself. Thus, changing the status of a disguised item does not change any test outcome by itself, because all tests containing it are already masked by other infected items.

The key observation is that each pair $(i, j) \in V_1^+(X, u^n) \times V_0^+(X, u^n)$ generates another infection vector with the same number of infected items and the same test outcome. Define v^n by removing the infected disguised item i and adding the

healthy disguised item j such that $v_i = 0, v_j = 1, v_\ell = u_\ell$ for all $\ell \notin \{i, j\}$. Then v^n has the same Hamming weight as u^n . Moreover, every test that contains i still remains positive because i was disguised, and every test that contains j was already positive because j was disguised. All other tests are unchanged. Therefore $Y(v^n) = Y(u^n)$. Consequently, if both $V_1^+(X, u^n)$ and $V_0^+(X, u^n)$ are large, then the decoder has many competing vectors that are indistinguishable from the true vector.

The following statement is the form of the Coja-Oghlan et al. [1] argument that we need.

Theorem 6: [Disguised individuals] For any $\epsilon > 0$, there exists $\theta_0 = \theta_0(\epsilon) < 1$ such that for every $\theta \in (\theta_0, 1)$, there exists $n_0 = n_0(\theta, \epsilon)$ such that for all $n > n_0$ and all non-adaptive test designs $X \in \{0, 1\}^{T \times n}$ with $\tau < \theta/\ln 2$, we have

$$\mathbb{P}(|V_0^+(X, U^n)| |V_1^+(X, U^n)| \geq \log n) > 1 - \epsilon.$$

Proof Sketch: The proof follows the disguised-individual argument of [1]. The only additional point is that the infection vector is sampled from a Markov source rather than uniformly from all k -subsets. For typical Markov sequences, however, the Hamming weight is $nq + o(nq)$ and the infected coordinates are sufficiently spread over the tests for the same masking calculation to apply. We consider this more precisely in a longer version of the paper.

The result above gives the disguised-individual lower bound for θ sufficiently close to 1. The reduction argument in [1] then extends this lower bound to all $\theta > \frac{\ln 2}{1 + \ln 2}$.

Therefore, throughout this regime, there is a non-vanishing probability that at least one other typical vector produces exactly the same test outcome as the true vector under any sequence of test matrices and decoders.

Combining the counting lower bound and the disguised-individual lower bound gives $\tau < \tau^*$ is not achievable. Together with the achievability result, this proves Theorem 3.

VI. CONCLUSION

We characterized the non-adaptive testing capacity under a Markovian infection prior. Although the main result is stated for a two-state Markov model, the proof techniques are more general. In particular, the same approach extends to any fixed k -th order Markov source, with the source-complexity term replaced by the corresponding sparsity-normalized entropy rate.

The proof relies on two ingredients: source-dependent typical-set counting and overlap-based indistinguishability bounds. The former controls the number of plausible competing infection vectors under the correlated prior, while the latter bounds the probability that a competitor with a given overlap produces the same test outcomes. Together, these yield a decoupling between the prior complexity and the test-design collision probability.

These ingredients are not specific to the two-state model. They provide a natural framework for broader correlated priors.

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APPENDIX

Lemma 3: [Expected $N(\lambda)$ bound] Fix $\epsilon > 0$. Then

$$\mathbb{E} \left[N(\lambda \mid U^n) \mid U^n \in T_\epsilon^{(n)} \right] \leq 2^{nq \log n [\beta(1-\theta)(1-\lambda) + \epsilon + o_n(1)]}.$$

Proof: To prove this lemma we first state a more general lemma below

Lemma 4: [$N(\lambda)$ general bound] Let U^n and V^n be independent and identically distributed according to P_{U^n} . Define

$$J_n(\lambda) := -\frac{1}{n} \log \mathbb{P}(l(U^n, V^n) = \lambda k \mid (U^n, V^n) \in T_\epsilon \times T_\epsilon). \quad (14)$$

Then

$$\mathbb{E} \left[N_\epsilon(\lambda \mid U^n) \mid U^n \in T_\epsilon^{(n)} \right] \leq \frac{1}{1 - \delta_n} 2^{n(\tilde{H}_n + 3\epsilon - J_n(\lambda)) + o(n)}. \quad (15)$$

This Lemma is proved in Appendix A. We first compute the leading-order entropy of the Markov typical set. The stationary distribution of the two-state Markov chain is

$$\pi_1 = q_n = \frac{\alpha_n}{\alpha_n + \beta}, \quad \pi_0 = 1 - q_n.$$

Hence the entropy rate is

$$H_n = H(U_2 \mid U_1) = (1 - q_n)h(\alpha_n) + q_n h(\beta),$$

where $h(\cdot)$ denotes the binary entropy function. Since

$$q_n = \Theta(n^{\theta-1}), \quad \alpha_n = \frac{\beta q_n}{1 - q_n} = \beta q_n(1 + o(1)),$$

we have

$$h(\alpha_n) = \alpha_n \log \frac{1}{\alpha_n} + O(\alpha_n),$$

and therefore

$$(1 - q_n)h(\alpha_n) = \alpha_n \log \frac{1}{\alpha_n} + O(\alpha_n).$$

Using $\alpha_n = \beta q_n(1 + o(1))$, this becomes

$$(1 - q_n)h(\alpha_n) = \beta q_n \log \frac{1}{q_n} + O(q_n).$$

The second term satisfies

$$q_n h(\beta) = O(q_n),$$

because $\beta \in (0, 1]$ is fixed. Hence

$$H_n = \beta q_n \log \frac{1}{q_n} + O(q_n).$$

Since

$$q_n = \Theta(n^{\theta-1}),$$

we have

$$\log \frac{1}{q_n} = (1 - \theta) \log n + O(1).$$

Therefore,

$$H_n = \beta q_n (1 - \theta) \log n + O(q_n).$$

Multiplying by n , the logarithmic size of the Markov typical set satisfies

$$n\tilde{H}_n = nH_n + o(nq_n \log n) = \beta nq_n (1 - \theta) \log n + o(nq_n \log n).$$

Equivalently,

$$\tilde{H}_n = \beta q_n (1 - \theta) \log n + o(q_n \log n).$$

We show that

$$J_n(\lambda) = \beta \lambda q_n (1 - \theta) \log n + o_\epsilon(q_n \log n) + o_n(q_n \log n).$$

For a sequence x^n , define

$$K(x^n) := \sum_{i=1}^n x_i,$$

and

$$R(x^n) := \sum_{i=1}^{n-1} \mathbf{1}\{x_i = 1, x_{i+1} = 0\}.$$

For $\eta > 0$, define

$$G_\eta^{(n)} := \{x^n : |R(x^n) - \beta K(x^n)| \leq \eta k\}.$$

We then define the refined typical set

$$\tilde{T}_{\epsilon, \eta}^{(n)} := T_\epsilon^{(n)} \cap G_\eta^{(n)}.$$

By concentration for bounded additive of finite-state Markov chains, for every fixed $\eta > 0$,

$$\mathbb{P}\left(U^n \notin G_\eta^{(n)}\right) \leq \exp\{-c_\eta k + o(k)\}$$

for some $c_\eta > 0$. Hence

$$\mathbb{P}\left(U^n \in \tilde{T}_{\epsilon, \eta}^{(n)}\right) \geq 1 - \delta_n - \exp\{-c_\eta k + o(k)\}.$$

In particular,

$$\mathbb{P}\left(U^n \in \tilde{T}_{\epsilon, \eta}^{(n)}\right) = 1 - o(1).$$

From this point onward, the typical set used in the overlap-counting argument is $\tilde{T}_{\epsilon, \eta}^{(n)}$. To simplify notation, we continue to write it as $T_\epsilon^{(n)}$. Thus, for every $x^n \in T_\epsilon^{(n)}$,

$$K(x^n) = k(1 + o_\epsilon(1)),$$

and

$$R(x^n) = \beta K(x^n) + O(\eta k) = \beta k + O(\eta k) + o_\epsilon(k).$$

Equivalently,

$$R(x^n) = (\beta + O(\eta) + o_\epsilon(1))k.$$

We now compute

$$J_n(\lambda) := -\frac{1}{n} \log \mathbb{P}\left(l(U^n, V^n) = \lambda k \mid U^n, V^n \in T_\epsilon^{(n)}\right),$$

where $T_\epsilon^{(n)}$ denotes the refined typical set above.

Fix $u^n \in T_\epsilon^{(n)}$, and let

$$S := S(u^n) := \{i : u_i = 1\}.$$

Then

$$|S| = K(u^n) = k(1 + o_\epsilon(1)).$$

For any $v^n \in T_\epsilon^{(n)}$, the number of infected runs of v^n is

$$R(v^n) = (\beta + O(\eta) + o_\epsilon(1))k.$$

Let

$$m_n(v^n) := R(v^n).$$

The run-start locations of v^n determine the leading $k \log n$ -scale entropy. The remaining choices, namely the run lengths and the $O(k)$ -scale internal configurations, contribute only

$$2^{O(k)} = 2^{o(k \log n)}$$

possibilities and hence do not affect the $k \log n$ -scale exponent.

The total number of admissible run-start configurations for a sequence $v^n \in T_\epsilon^{(n)}$ is therefore

$$\binom{n}{(\beta + O(\eta) + o_\epsilon(1))k} 2^{o(k \log n)}.$$

Now impose the overlap condition

$$l(u^n, v^n) = \lambda k.$$

Since $v^n \in T_\epsilon^{(n)}$, its empirical average infected-run length is

$$\frac{K(v^n)}{R(v^n)} = \frac{k(1 + o_\epsilon(1))}{(\beta + O(\eta) + o_\epsilon(1))k} = \frac{1}{\beta + O(\eta) + o_\epsilon(1)}.$$

Hence an overlap of λk infected coordinates corresponds to

$$(\beta\lambda + O(\eta) + o_\epsilon(1))k$$

infected runs intersecting S . Therefore, the number of admissible run-start configurations compatible with the overlap constraint is

$$\binom{k(1 + o_\epsilon(1))}{(\beta\lambda + O(\eta) + o_\epsilon(1))k} \binom{n - k(1 + o_\epsilon(1))}{(\beta(1 - \lambda) + O(\eta) + o_\epsilon(1))k} 2^{o(k \log n)}.$$

Thus, uniformly over $u^n \in T_\epsilon^{(n)}$,

$$\begin{aligned} \mathbb{P} \left(l(u^n, V^n) = \lambda k \mid V^n \in T_\epsilon^{(n)} \right) \\ = \frac{\binom{k(1 + o_\epsilon(1))}{(\beta\lambda + O(\eta) + o_\epsilon(1))k} \binom{n - k(1 + o_\epsilon(1))}{(\beta(1 - \lambda) + O(\eta) + o_\epsilon(1))k}}{\binom{n}{(\beta + O(\eta) + o_\epsilon(1))k}} 2^{o(k \log n)}. \end{aligned} \quad (16)$$

Taking negative logarithms gives

$$\begin{aligned} -\log \mathbb{P} \left(l(u^n, V^n) = \lambda k \mid V^n \in T_\epsilon^{(n)} \right) \\ = \log \binom{n}{(\beta + O(\eta) + o_\epsilon(1))k} \\ - \log \binom{k(1 + o_\epsilon(1))}{(\beta\lambda + O(\eta) + o_\epsilon(1))k} \\ - \log \binom{n - k(1 + o_\epsilon(1))}{(\beta(1 - \lambda) + O(\eta) + o_\epsilon(1))k} + o(k \log n). \end{aligned} \quad (17)$$

Since $k = o(n)$, Stirling's approximation gives

$$\log \binom{n}{(\beta + O(\eta) + o_\epsilon(1))k} = (\beta + O(\eta) + o_\epsilon(1))k \log \frac{n}{k} + O(k),$$

and

$$\log \binom{n - k(1 + o_\epsilon(1))}{(\beta(1 - \lambda) + O(\eta) + o_\epsilon(1))k} = (\beta(1 - \lambda) + O(\eta) + o_\epsilon(1))k \log \frac{n}{k} + O(k).$$

Moreover,

$$\log \binom{k(1 + o_\epsilon(1))}{(\beta\lambda + O(\eta) + o_\epsilon(1))k} = O(k).$$

Therefore,

$$\begin{aligned} -\log \mathbb{P} \left(l(u^n, V^n) = \lambda k \mid V^n \in T_\epsilon^{(n)} \right) \\ = [\beta - \beta(1 - \lambda) + O(\eta) + o_\epsilon(1)] k \log \frac{n}{k} + o(k \log n) \\ = [\beta\lambda + O(\eta) + o_\epsilon(1)] k \log \frac{n}{k} + o(k \log n). \end{aligned} \quad (18)$$

Since

$$k = nq_n(1 + o(1)),$$

we have

$$\log \frac{n}{k} = \log \frac{1}{q_n} + o(1).$$

Since

$$q_n = \Theta(n^{\theta-1}),$$

we also have

$$\log \frac{1}{q_n} = (1 - \theta) \log n + O(1).$$

Hence

$$\begin{aligned} & -\log \mathbb{P} \left(l(u^n, V^n) = \lambda k \mid V^n \in T_\epsilon^{(n)} \right) \\ & = [\beta \lambda + O(\eta) + o_\epsilon(1) + o_n(1)] n q_n (1 - \theta) \log n. \end{aligned} \quad (19)$$

The estimate is uniform over $u^n \in T_\epsilon^{(n)}$. Averaging over the conditional law of U^n on $T_\epsilon^{(n)}$, we obtain

$$\begin{aligned} & -\log \mathbb{P} \left(l(U^n, V^n) = \lambda k \mid U^n, V^n \in T_\epsilon^{(n)} \right) \\ & = [\beta \lambda + O(\eta) + o_\epsilon(1) + o_n(1)] n q_n (1 - \theta) \log n. \end{aligned} \quad (20)$$

Dividing by n ,

$$J_n(\lambda) = [\beta \lambda + O(\eta) + o_\epsilon(1) + o_n(1)] q_n (1 - \theta) \log n.$$

Finally, sending $\eta \downarrow 0$ after $n \rightarrow \infty$ and $\epsilon \downarrow 0$, we obtain

$$J_n(\lambda) = \beta \lambda q_n (1 - \theta) \log n + o_\epsilon(q_n \log n) + o_n(q_n \log n).$$

Theorem 4: [Small-overlap probability] Fix $u^n, v^n \in T_\epsilon^{(n)}$ with $\lambda(u^n, v^n) = \lambda \leq \lambda_n^*$. Under the above test design,

$$\mathbb{P}(Y(v^n) = Y(u^n) \mid \mathcal{O}_\lambda(u^n, v^n)) \leq 2^{-T(1-\lambda)[1-\epsilon-o_n(1)]}.$$

Proof: This follows the steps in [9]. We start with the bound on the collision probability obtained under the mild dependencies generated by the constant column case.

Lemma 5 (Bound on the collision probability): Let

$$p(\lambda \mid u^n) := \mathbb{P} \left(Y(v^n) = Y(u^n) \mid u^n, v^n \in T_\epsilon^{(n)}, \text{ overlap } \lambda \right),$$

where $\ell = \lambda k$. Define

$$\Delta := \frac{\nu T}{k}, \quad \mu := \frac{\Delta n}{T}, \quad d_n := \sqrt{\frac{\Delta n}{T}} \log n,$$

and

$$P_T^+(\lambda) := 1 - 2 \left(1 - \frac{k + \epsilon}{n} \right)^{\mu + d_n} + 2 \left(1 - \frac{2k - \ell - 2\epsilon}{n} \right)^{\mu - d_n}.$$

Then, for any fixed pair $u^n, v^n \in T_\epsilon^{(n)}$ with overlap $\ell = \lambda k$, we have

$$p(\lambda \mid u^n) \leq C (\Delta(k + \epsilon))^{3/2} \left[P_T^+(\lambda)^T + o(n^{-2}) \right].$$

For the remainder of the argument, we work implicitly on the event

$$\{u^n, v^n \in T_\epsilon^{(n)}\}.$$

Recall that

$$K(u^n) := \{i : u_i = 1\}, \quad K(v^n) := \{i : v_i = 1\},$$

and define

$$k_u := |K(u^n)|, \quad k_v := |K(v^n)|, \quad \ell := |K(u^n) \cap K(v^n)|.$$

Also write

$$s := |K(u^n) \setminus K(v^n)| = k_u - \ell, \quad t := |K(v^n) \setminus K(u^n)| = k_v - \ell.$$

Thus

$$|K(u^n) \cup K(v^n)| = k_u + k_v - \ell.$$

Recall that each item chooses Δ tests, and for each test $r \in \{1, \dots, T\}$,

$$\Gamma_r := \sum_{i=1}^n N_{r,i},$$

where $N_{r,i}$ is the number of times item i selects test r among its Δ draws. Let

$$\Gamma := \sigma(\Gamma_1, \dots, \Gamma_T),$$

the sigma algebra generated by Γ_i for $i \in [T]$. Conditioned on Γ , the testing matrix may be generated by the pairing model [1]: Create Δ clones of each item $i \in [n]$ and Γ_r clones of each test $r \in [T]$, and choose a uniformly random perfect matching (Every item-clone and test-clone have a unique mapping) between the Δn item-clones and the $\sum_{r=1}^T \Gamma_r = \Delta n$ test-clones. Each matched pair between a clone of item i and a clone of test r contributes an i - r edge. This produces exactly the near-constant-column design conditioned on Γ .

We are interested in the collision event

$$\mathcal{S} := \{Y(v^n) = Y(u^n)\} = \bigcap_{r=1}^T \{Y_r(u^n) = Y_r(v^n)\}.$$

We now introduce an auxiliary random variable. Conditioned on Γ , let

$$(B_{r,j})_{r \in [T], j \in [\Gamma_r]}$$

be mutually independent draws taking values in $[n]$, each uniformly distributed on $[n]$. From these draws define the artificial test outcomes

$$Y'_r(u^n) := \max_{j \in [\Gamma_r]} 1\{B_{r,j} \in K(u^n)\}, \quad Y'_r(v^n) := \max_{j \in [\Gamma_r]} 1\{B_{r,j} \in K(v^n)\},$$

and let

$$\mathcal{A} := \bigcap_{r=1}^T \{Y'_r(u^n) = Y'_r(v^n)\}.$$

The variables $B_{r,j}$ should be interpreted as an auxiliary “balls-into-bins” representation of the test design after conditioning on the row degrees $\Gamma = (\Gamma_1, \dots, \Gamma_T)$. For each test r , the value Γ_r specifies how many item selections fall into that test. The auxiliary variables $B_{r,1}, \dots, B_{r,\Gamma_r}$ then assign these selections independently and uniformly to items in $[n]$.

Thus, $Y'_r(u^n)$ records whether at least one of the Γ_r selections in test r hits the support $K(u^n)$, and $Y'_r(v^n)$ is defined analogously for $K(v^n)$. The event $\mathcal{A}$ is therefore the collision event in this auxiliary model: for every test r , either both supports are hit or neither support is hit. This auxiliary construction is useful because, conditioned on Γ , the tests become independent across r .

Next define the event $\mathcal{T}$ that, across the full collection of draws $(B_{r,j})$, each item $i \in [n]$ is sampled exactly Δ times. Equivalently,

$$\mathcal{T} = \left\{ \sum_{r=1}^T \sum_{j=1}^{\Gamma_r} 1\{B_{r,j} = i\} = \Delta \text{ for every } i \in [n] \right\}.$$

Since $\sum_{r=1}^T \Gamma_r = \Delta n$, this event is consistent. Conditioned on Γ and $\mathcal{T}$, the array $(B_{r,j})$ is uniformly distributed over all assignments of item labels to the Δn test-clones such that each item appears exactly Δ times. But this is exactly the pairing-model law. Hence

$$(Y_r(u^n), Y_r(v^n))_{r=1}^T \mid \Gamma \stackrel{d}{=} (Y'_r(u^n), Y'_r(v^n))_{r=1}^T \mid \Gamma, \mathcal{T},$$

and therefore

$$\mathbb{P}(\mathcal{S} \mid \Gamma) = \mathbb{P}(\mathcal{A} \mid \Gamma, \mathcal{T}).$$

Since the variables $(B_{r,j})$ are independent, the rows are independent in the auxiliary experiment, and so

$$\mathbb{P}(\mathcal{A} \mid \Gamma) = \prod_{r=1}^T \mathbb{P}(Y'_r(u^n) = Y'_r(v^n) \mid \Gamma). \quad (21)$$

Now fix a row r . Since

$$\{Y'_r(u^n) = Y'_r(v^n)\} = \{Y'_r(u^n) = 0, Y'_r(v^n) = 0\} \cup \{Y'_r(u^n) = 1, Y'_r(v^n) = 1\},$$

we have

$$\begin{aligned} \mathbb{P}(Y'_r(u^n) = Y'_r(v^n) \mid \Gamma) &= 1 - \mathbb{P}(Y'_r(u^n) = 0 \mid \Gamma) - \mathbb{P}(Y'_r(v^n) = 0 \mid \Gamma) \\ &\quad + 2\mathbb{P}(Y'_r(u^n) = 0, Y'_r(v^n) = 0 \mid \Gamma). \end{aligned}$$

Moreover, by independence of the Γ_r draws in row r ,

$$\begin{aligned}\mathbb{P}(Y'_r(u^n) = 0 \mid \Gamma) &= \left(1 - \frac{k_u}{n}\right)^{\Gamma_r}, \\ \mathbb{P}(Y'_r(v^n) = 0 \mid \Gamma) &= \left(1 - \frac{k_v}{n}\right)^{\Gamma_r}, \\ \mathbb{P}(Y'_r(u^n) = 0, Y'_r(v^n) = 0 \mid \Gamma) &= \left(1 - \frac{k_u + k_v - \ell}{n}\right)^{\Gamma_r}.\end{aligned}$$

Hence

$$\mathbb{P}(\mathcal{A} \mid \Gamma) = \prod_{r=1}^T \left[1 - \left(1 - \frac{k_u}{n}\right)^{\Gamma_r} - \left(1 - \frac{k_v}{n}\right)^{\Gamma_r} + 2 \left(1 - \frac{k_u + k_v - \ell}{n}\right)^{\Gamma_r} \right]. \quad (22)$$

Next, by Bayes' rule,

$$\mathbb{P}(\mathcal{S} \mid \Gamma) = \mathbb{P}(\mathcal{A} \mid \Gamma, \mathcal{T}) = \frac{\mathbb{P}(\mathcal{A} \mid \Gamma) \mathbb{P}(\mathcal{T} \mid \mathcal{A}, \Gamma)}{\mathbb{P}(\mathcal{T} \mid \Gamma)} \leq \frac{\mathbb{P}(\mathcal{A} \mid \Gamma)}{\mathbb{P}(\mathcal{T} \mid \Gamma)}. \quad (23)$$

It remains to lower bound $\mathbb{P}(\mathcal{T} \mid \Gamma)$. We condition only on the aggregate counts induced by the partition of $[n]$ determined by $K(u^n)$ and $K(v^n)$. Namely,

$$\begin{aligned}K(u^n) \cap K(v^n), \quad K(u^n) \setminus K(v^n), \quad K(v^n) \setminus K(u^n), \quad (K(u^n) \cup K(v^n))^c, \\ \ell := |K(u^n) \cap K(v^n)|, \quad k_u := |K(u^n)|, \quad k_v := |K(v^n)|, \\ p_{11} := \frac{\ell}{n}, \quad p_{10} := \frac{k_u - \ell}{n}, \quad p_{01} := \frac{k_v - \ell}{n}, \quad p_{00} := 1 - \frac{k_u + k_v - \ell}{n}.\end{aligned}$$

Here $p_{11}, p_{10}, p_{01}, p_{00}$ are the probabilities that a single auxiliary draw $B_{r,j}$ falls in the four categories above, respectively. Conditioned on Γ , the total number of auxiliary draws is

$$\sum_{r=1}^T \Gamma_r = \Delta n.$$

Let

$$Z := (Z_{11}, Z_{10}, Z_{01}, Z_{00})$$

denote the number of auxiliary draws falling into the four categories above. Then

$$Z \sim \text{Mult}(\Delta n; p_{11}, p_{10}, p_{01}, p_{00}).$$

We define $\mathcal{T}$ to be the event that these four aggregate counts equal their expected values, namely

$$\mathcal{T} := \{Z_{11} = \Delta \ell, Z_{10} = \Delta(k_u - \ell), Z_{01} = \Delta(k_v - \ell), Z_{00} = \Delta(n - k_u - k_v + \ell)\}.$$

Thus

$$\begin{aligned}\mathbb{P}(\mathcal{T} \mid \Gamma) &= \frac{(\Delta n)!}{(\Delta \ell)! (\Delta(k_u - \ell))! (\Delta(k_v - \ell))! (\Delta(n - k_u - k_v + \ell))!} \\ &\quad \times \left(\frac{\ell}{n}\right)^{\Delta \ell} \left(\frac{k_u - \ell}{n}\right)^{\Delta(k_u - \ell)} \left(\frac{k_v - \ell}{n}\right)^{\Delta(k_v - \ell)} \left(1 - \frac{k_u + k_v - \ell}{n}\right)^{\Delta(n - k_u - k_v + \ell)}.\end{aligned} \quad (24)$$

Using Stirling's formula in the form

$$m! = \Theta\left(\sqrt{m} \left(\frac{m}{e}\right)^m\right),$$

the exponential terms cancel because $\mathcal{T}$ is the event that the multinomial vector hits its mean. Hence

$$\begin{aligned}\mathbb{P}(\mathcal{T} \mid \Gamma) &= \Theta\left(\frac{\sqrt{\Delta n}}{\sqrt{\Delta \ell} \sqrt{\Delta(k_u - \ell)} \sqrt{\Delta(k_v - \ell)} \sqrt{\Delta(n - k_u - k_v + \ell)}}\right) \\ &= \Theta\left(\frac{\sqrt{n}}{\Delta^{3/2} \sqrt{\ell(k_u - \ell)(k_v - \ell)(n - k_u - k_v + \ell)}}\right).\end{aligned} \quad (25)$$

Since $k_u, k_v \leq k + \epsilon$ and $n - k_u - k_v + \ell \leq n$, we have

$$\ell \leq k + \epsilon, \quad k_u - \ell \leq k + \epsilon, \quad k_v - \ell \leq k + \epsilon, \quad n - k_u - k_v + \ell \leq n.$$

Therefore,

$$\sqrt{\ell (k_u - \ell) (k_v - \ell) (n - k_u - k_v + \ell)} \leq \sqrt{n} (k + \epsilon)^{3/2}.$$

Substituting this into (25), there exists a constant $c > 0$ such that

$$\mathbb{P}(\mathcal{T} \mid \Gamma) \geq c (\Delta (k + \epsilon))^{-3/2}. \quad (26)$$

Consequently,

$$\frac{1}{\mathbb{P}(\mathcal{T} \mid \Gamma)} \leq C (\Delta (k + \epsilon))^{3/2}. \quad (27)$$

Since $\Delta = \Theta(\log n)$ and $k = n^\theta$, this prefactor is polynomial in k and Δ , and is therefore negligible on the exponential scale $k \log n = n^\theta \log n$.

Therefore, combining (22), (23), and (27), we obtain

$$\begin{aligned} \mathbb{P}(Y(v^n) = Y(u^n) \mid u^n, v^n, \Gamma) \\ \leq C (\Delta (k + \epsilon))^{3/2} \prod_{r=1}^T \left[1 - \left(1 - \frac{k_u}{n}\right)^{\Gamma_r} - \left(1 - \frac{k_v}{n}\right)^{\Gamma_r} + 2 \left(1 - \frac{k_u + k_v - \ell}{n}\right)^{\Gamma_r} \right]. \end{aligned} \quad (28)$$

We now use the typical-set bounds

$$k - \epsilon \leq k_u \leq k + \epsilon, \quad k - \epsilon \leq k_v \leq k + \epsilon.$$

For each r , the bracket in (28) is

$$1 - \left(1 - \frac{k_u}{n}\right)^{\Gamma_r} - \left(1 - \frac{k_v}{n}\right)^{\Gamma_r} + 2 \left(1 - \frac{k_u + k_v - \ell}{n}\right)^{\Gamma_r}.$$

Since the first two nonconstant terms appear with negative signs, replacing k_u and k_v by the upper bound $k + \epsilon$ gives an upper bound. Similarly, since the final term appears with a positive sign, replacing $k_u + k_v - \ell$ by the lower bound

$$k_u + k_v - \ell \geq 2k - \ell - 2\epsilon$$

also gives an upper bound. Therefore, provided $2k - \ell - 2\epsilon \geq 0$, we have

$$\begin{aligned} \mathbb{P}(Y(v^n) = Y(u^n) \mid u^n, v^n, \Gamma) \\ \leq C (\Delta (k + \epsilon))^{3/2} \prod_{r=1}^T \left[1 - 2 \left(1 - \frac{k + \epsilon}{n}\right)^{\Gamma_r} + 2 \left(1 - \frac{2k - \ell - 2\epsilon}{n}\right)^{\Gamma_r} \right]. \end{aligned} \quad (29)$$

Taking expectation over Γ then yields

$$\begin{aligned} \mathbb{P}(Y(v^n) = Y(u^n) \mid u^n, v^n) \\ \leq C (\Delta (k + \epsilon))^{3/2} \mathbb{E}_\Gamma \left[\prod_{r=1}^T \left(1 - 2 \left(1 - \frac{k + \epsilon}{n}\right)^{\Gamma_r} + 2 \left(1 - \frac{2k - \ell - 2\epsilon}{n}\right)^{\Gamma_r} \right) \right]. \end{aligned} \quad (30)$$

Lemma 6 (Uniform concentration of row degrees): Let Γ_r denote the number of item selections assigned to test $r \in [T]$ under the near-constant-column design, where each of the n items selects Δ tests independently and uniformly from $[T]$. Define

$$\Gamma_{\min} := \min_{r \in [T]} \Gamma_r, \quad \Gamma_{\max} := \max_{r \in [T]} \Gamma_r.$$

Assume that $T \leq n^C$ for some constant $C > 0$ and that

$$\frac{\Delta n}{T} \geq c \log^2 n$$

for some constant $c > 0$. Then, with probability at least $1 - o(n^{-2})$,

$$\frac{\Delta n}{T} - \sqrt{\frac{\Delta n}{T}} \log n \leq \Gamma_{\min} \leq \Gamma_{\max} \leq \frac{\Delta n}{T} + \sqrt{\frac{\Delta n}{T}} \log n.$$

For each test $r \in [T]$, Γ_r is the number of times test r is selected among the total Δn item-test selections. Hence

$$\Gamma_r \sim \text{Bin}\left(\Delta n, \frac{1}{T}\right),$$

and therefore

$$\mathbb{E}[\Gamma_r] = \frac{\Delta n}{T}.$$

Let

$$\mu := \frac{\Delta n}{T} \sim O(n^{1-\theta}).$$

We apply a standard Chernoff bound. For any $a > 0$,

$$\mathbb{P}(|\Gamma_r - \mu| \geq a) \leq 2 \exp\left(-\frac{a^2}{2(\mu + a/3)}\right).$$

Choose

$$a := \sqrt{\mu} \log n.$$

Since for any $\theta < 1$, $\mu \geq c \log^2 n$, we have $a \leq C' \mu$ for some constant $C' > 0$. Hence

$$\frac{a^2}{2(\mu + a/3)} = \frac{\mu \log^2 n}{2(\mu + \sqrt{\mu} \log n/3)} \geq c' \log^2 n$$

for some constant $c' > 0$. Therefore, for each fixed $r \in [T]$,

$$\mathbb{P}(|\Gamma_r - \mu| \geq \sqrt{\mu} \log n) \leq 2 \exp(-c' \log^2 n).$$

Applying the union bound over all $r \in [T]$ gives

$$\mathbb{P}(\exists r \in [T] : |\Gamma_r - \mu| \geq \sqrt{\mu} \log n) \leq 2T \exp(-c' \log^2 n). \quad (31)$$

Since $T \leq n^C$, the right-hand side satisfies

$$2T \exp(-c' \log^2 n) \leq 2 \exp(C \log n - c' \log^2 n) = o(n^{-2}).$$

Thus, with probability at least $1 - o(n^{-2})$, for every $r \in [T]$,

$$\left|\Gamma_r - \frac{\Delta n}{T}\right| \leq \sqrt{\frac{\Delta n}{T}} \log n.$$

Equivalently,

$$\frac{\Delta n}{T} - \sqrt{\frac{\Delta n}{T}} \log n \leq \Gamma_r \leq \frac{\Delta n}{T} + \sqrt{\frac{\Delta n}{T}} \log n$$

simultaneously for all $r \in [T]$. Taking the minimum and maximum over $r \in [T]$ gives

$$\frac{\Delta n}{T} - \sqrt{\frac{\Delta n}{T}} \log n \leq \Gamma_{\min} \leq \Gamma_{\max} \leq \frac{\Delta n}{T} + \sqrt{\frac{\Delta n}{T}} \log n,$$

as claimed.

Let

$$\mu := \frac{\Delta n}{T}, \quad d_n := \sqrt{\frac{\Delta n}{T}} \log n.$$

By the row-degree concentration lemma, with probability at least $1 - o(n^{-2})$,

$$\Gamma_{\min} \geq \mu - d_n, \quad \Gamma_{\max} \leq \mu + d_n.$$

Define

$$a_n := 1 - \frac{k + \epsilon}{n}, \quad b_n := 1 - \frac{2k - \ell - 2\epsilon}{n}.$$

Since $0 \leq b_n \leq a_n \leq 1$, for every $r \in [T]$ on this event we have

$$a_n^{\Gamma_r} \geq a_n^{\mu + d_n}, \quad b_n^{\Gamma_r} \leq b_n^{\mu - d_n}.$$

Therefore,

$$1 - 2a_n^{\Gamma_r} + 2b_n^{\Gamma_r} \leq 1 - 2a_n^{\mu+d_n} + 2b_n^{\mu-d_n}. \quad (32)$$

It follows that, on the same event,

$$\prod_{r=1}^T (1 - 2a_n^{\Gamma_r} + 2b_n^{\Gamma_r}) \leq [1 - 2a_n^{\mu+d_n} + 2b_n^{\mu-d_n}]^T. \quad (33)$$

On the complement of this event, we use the trivial bound that the product is at most one. Hence

$$\begin{aligned} & \mathbb{E}_\Gamma \left[\prod_{r=1}^T \left(1 - 2 \left(1 - \frac{k+\epsilon}{n} \right)^{\Gamma_r} + 2 \left(1 - \frac{2k-\ell-2\epsilon}{n} \right)^{\Gamma_r} \right) \right] \\ & \leq \left[1 - 2 \left(1 - \frac{k+\epsilon}{n} \right)^{\mu+d_n} + 2 \left(1 - \frac{2k-\ell-2\epsilon}{n} \right)^{\mu-d_n} \right]^T + o(n^{-2}). \end{aligned} \quad (34)$$

Substituting this into (30) gives the closed-form bound

$$\begin{aligned} & \mathbb{P}(Y(v^n) = Y(u^n) \mid u^n, v^n) \\ & \leq C(\Delta(k+\epsilon))^{3/2} \left[\left(1 - 2 \left(1 - \frac{k+\epsilon}{n} \right)^{\mu+d_n} + 2 \left(1 - \frac{2k-\ell-2\epsilon}{n} \right)^{\mu-d_n} \right)^T + o(n^{-2}) \right]. \end{aligned} \quad (35)$$

Theorem 5: [Large-overlap error bound] Let $\lambda_n^* = 1 - 1/\log n$ and choose $\nu = \ln 2$. For every $u^n \in T_\epsilon^{(n)}$,

$$\mathbb{P}(\mathcal{C}_{>\lambda_n^*}(u^n) \mid U^n = u^n) \leq n^{\theta - \frac{T \ln 2}{nq \log n} + o(1)}.$$

Proof: This follows the proof in [1]

Theorem 6: [Disguised individuals] For any $\epsilon > 0$, there exists $\theta_0 = \theta_0(\epsilon) < 1$ such that for every $\theta \in (\theta_0, 1)$, there exists $n_0 = n_0(\theta, \epsilon)$ such that for all $n > n_0$ and all non-adaptive test designs $\mathbf{X} \in \{0, 1\}^{T \times n}$ with $\tau < \theta/\ln 2$, we have

$$\mathbb{P}(|V_0^+(\mathbf{X}, U^n)| |V_1^+(\mathbf{X}, U^n)| \geq \log n) > 1 - \epsilon.$$

Fix $\epsilon > 0$. There exists $\theta_0 = \theta_0(\epsilon) < 1$ such that for every $\theta \in (\theta_0, 1)$ there exists $n_0 = n_0(\theta, \epsilon)$ such that the following holds for all $n \geq n_0$.

Let $X^n = (X_1, \dots, X_n)$ be the stationary two-state Markov chain with transition matrix

$$P_n = \begin{pmatrix} 1 - \alpha_n & \alpha_n \\ \beta & 1 - \beta \end{pmatrix}, \quad q_n = \mathbb{P}(X_i = 1) = \frac{\alpha_n}{\alpha_n + \beta},$$

where $\beta \in (0, 1]$ is fixed and

$$q_n = \Theta(n^{\theta-1}).$$

Let

$$k_n := nq_n.$$

For any test design $G = (V_n \cup F_m, E)$ with $V_n = [n]$ and

$$m \leq (1 - \epsilon) \frac{\theta}{(\ln 2)^2} k_n \ln n,$$

let

$$V_+(G, X^n) := \{x \in V_n : \forall a \in \partial x, \exists y \in \partial a \setminus \{x\} \text{ such that } X_y = 1\}.$$

Also define

$$V_{0,+}(G, X^n) := \{x \in V_+(G, X^n) : X_x = 0\},$$

and

$$V_{1,+}(G, X^n) := \{x \in V_+(G, X^n) : X_x = 1\}.$$

Then

$$\mathbb{P}(|V_{0,+}(G, X^n)| \wedge |V_{1,+}(G, X^n)| \geq \ln n) \geq 1 - \epsilon.$$

Proof: Let

$$\Gamma := \frac{\ln n}{q_n}.$$

We first remove tests of degree larger than Γ . Let G^* be the test design obtained from G by deleting all tests $a \in F_m$ satisfying

$$|\partial a| > \Gamma.$$

We show that, with probability $1 - o(1)$, deleting these tests does not change the disguised status of any individual.

For any set $A \subseteq [n]$, write

$$X_A := \sum_{i \in A} X_i.$$

Since the Markov chain is stationary and has marginal q_n ,

$$\mathbb{E}[X_A] = q_n |A|.$$

Moreover, the two-state chain is exponentially mixing at rate $1 - \alpha_n - \beta$, and β is fixed. Hence there exists a constant $c > 0$, depending only on β , such that for every $A \subseteq [n]$ and every $t > 0$,

$$\mathbb{P}\left(X_A \leq \frac{1}{2} q_n |A|\right) \leq \exp\{-c q_n |A|\}.$$

Therefore, if $|A| > \Gamma$, then

$$\mathbb{P}(X_A \leq 1) \leq \mathbb{P}\left(X_A \leq \frac{1}{2} q_n |A|\right) \leq \exp\{-c q_n |A|\} \leq n^{-c+o(1)}.$$

Taking n_0 large enough, and increasing the constant in the definition of Γ if necessary, we may assume that

$$\mathbb{P}(\exists a \in F_m : |\partial a| > \Gamma, |\partial a \cap S(X^n)| \leq 1) = o(1),$$

where

$$S(X^n) := \{i : X_i = 1\}.$$

On the complement of this event, every deleted test contains at least two infected individuals. Consequently,

$$V_{0,+}(G, X^n) = V_{0,+}(G^*, X^n), \quad V_{1,+}(G, X^n) = V_{1,+}(G^*, X^n).$$

Next we remove individuals of very large test-degree. Let

$$G^{(0)}$$

be the design obtained from G^* by deleting all individuals x with

$$|\partial_{G^*} x| > \ln^3 n.$$

Since every remaining test in G^* has degree at most Γ ,

$$\sum_{x \in V_n} |\partial_{G^*} x| = \sum_{a \in F(G^*)} |\partial_{G^*} a| \leq m\Gamma.$$

Using

$$m \leq (1 - \epsilon) \frac{\theta}{(\ln 2)^2} k_n \ln n$$

and

$$\Gamma = \frac{\ln n}{q_n}, \quad k_n = n q_n,$$

we obtain

$$m\Gamma \leq C n \ln^2 n$$

for a constant $C = C(\theta, \epsilon, \beta)$. Hence

$$|\{x \in V_n : |\partial_{G^*} x| > \ln^3 n\}| \leq \frac{C n}{\ln n}.$$

Thus

$$|V(G^{(0)})| \geq n - \frac{C n}{\ln n}.$$

Furthermore,

$$\max_{a \in F(G^{(0)})} |\partial_{G^{(0)}} a| \leq \Gamma, \quad \max_{x \in V(G^{(0)})} |\partial_{G^{(0)}} x| \leq \ln^3 n.$$

Choose $\xi = \xi(\epsilon, \theta) > 0$ such that

$$2(1 - \theta) < \xi < \epsilon\theta.$$

This is possible by taking θ sufficiently close to 1. Let

$$N := \lceil n^{1-\xi} \rceil.$$

We construct individuals $y_1, \dots, y_N$ inductively. Given $G^{(i-1)}$, choose $y_i \in V(G^{(i-1)})$ maximizing

$$\mathbb{P}(y \in V_+(G^{(i-1)}, X^n))$$

over $y \in V(G^{(i-1)})$. Then obtain $G^{(i)}$ from $G^{(i-1)}$ by deleting y_i and all vertices at graph distance at most 4 from y_i .

Since the maximum test-degree is at most Γ and the maximum individual-degree is at most $\ln^3 n$, each step removes at most

$$\Gamma^2 \ln^6 n$$

individuals. Therefore,

$$|V(G^{(i)})| \geq n - \frac{Cn}{\ln n} - i\Gamma^2 \ln^6 n.$$

Now

$$N\Gamma^2 \ln^6 n \leq n^{1-\xi} \cdot q_n^{-2} \ln^8 n.$$

Since $q_n = \Theta(n^{\theta-1})$,

$$n^{1-\xi} q_n^{-2} \ln^8 n = n^{1-\xi+2(1-\theta)} \ln^8 n = o(n),$$

because $\xi > 2(1 - \theta)$. Hence, uniformly for $0 \leq i \leq N$,

$$|V(G^{(i)})| = n(1 - o(1)).$$

For $x \in V(G^{(i)})$, let

$$D_x^{(i)} := \{x \in V_+(G^{(i)}, X^n)\}.$$

For $a \in \partial_{G^{(i)}} x$, define

$$D_{x,a}^{(i)} := \{\exists y \in \partial_{G^{(i)}} a \setminus \{x\} : X_y = 1\}.$$

Then

$$D_x^{(i)} = \bigcap_{a \in \partial_{G^{(i)}} x} D_{x,a}^{(i)}.$$

The events $D_{x,a}^{(i)}$ are increasing events in the coordinates of X^n . The stationary two-state Markov chain with transition matrix P_n is positively associated; hence the FKG inequality gives

$$\mathbb{P}(D_x^{(i)}) \geq \prod_{a \in \partial_{G^{(i)}} x} \mathbb{P}(D_{x,a}^{(i)}).$$

For a test a , put $d_a := |\partial_{G^{(i)}} a|$. Since

$$D_{x,a}^{(i)} = \left\{ \sum_{y \in \partial a \setminus \{x\}} X_y \geq 1 \right\},$$

and since the Markov chain has marginal q_n , the same block-concentration bound used above implies that there is a constant $c_1 = c_1(\beta) > 0$ such that

$$\mathbb{P}(D_{x,a}^{(i)}) \geq 1 - \exp\{-c_1 q_n (d_a - 1)\}.$$

Consequently,

$$\log \mathbb{P}(D_x^{(i)}) \geq \sum_{a \in \partial_{G^{(i)}} x} \log(1 - \exp\{-c_1 q_n (d_a - 1)\}).$$

Average this inequality over $x \in V(G^{(i)})$. Let

$$L_i := \frac{1}{|V(G^{(i)})|} \sum_{x \in V(G^{(i)})} \log \mathbb{P}(D_x^{(i)}).$$

Then

$$L_i \geq \frac{1}{|V(G^{(i)})|} \sum_{a \in F(G^{(i)})} d_a \log(1 - \exp\{-c_1 q_n(d_a - 1)\}). \quad (36)$$

The function

$$z \mapsto z \log(1 - \exp\{-c_1 q_n(z - 1)\})$$

is minimized, up to a multiplicative $1 + o(1)$ error, at

$$z = \frac{\ln 2}{c_1 q_n}.$$

Therefore there is a constant $C_\beta > 0$, depending only on β , such that

$$d_a \log(1 - \exp\{-c_1 q_n(d_a - 1)\}) \geq -\frac{C_\beta}{q_n}$$

for every a . Thus

$$L_i \geq -\frac{C_\beta |F(G^{(i)})|}{q_n |V(G^{(i)})|} \geq -\frac{(1 + o(1))C_\beta m}{n q_n}.$$

Using $k_n = n q_n$ and

$$m \leq (1 - \epsilon) \frac{\theta}{(\ln 2)^2} k_n \ln n,$$

we obtain

$$L_i \geq -(1 - \epsilon/2)\theta \ln n$$

for all $0 \leq i \leq N$, after decreasing ϵ and increasing n_0 if necessary.

Since y_i maximizes $\mathbb{P}(D_x^{(i-1)})$, we have

$$\log \mathbb{P}(D_{y_i}^{(i-1)}) \geq L_{i-1}.$$

Therefore,

$$\mathbb{P}(D_{y_i}^{(i-1)}) \geq n^{-(1-\epsilon/2)\theta}.$$

Because deleting tests and individuals can only make the disguised event harder, and because $G^{(i-1)}$ is an induced subdesign of G^* , we have

$$D_{y_i}^{(i-1)} \subseteq \{y_i \in V_+(G^*, X^n)\}.$$

Hence

$$\mathbb{P}(y_i \in V_+(G^*, X^n)) \geq n^{-(1-\epsilon/2)\theta}.$$

We now separate the disguised event from the infection status of y_i . The event

$$\{y_i \in V_+(G^*, X^n)\}$$

depends only on coordinates X_z with $z \neq y_i$. For the two-state Markov chain, the one-site conditional probabilities satisfy the finite-energy bounds

$$\mathbb{P}(X_i = 1 \mid X_{[n] \setminus \{i\}}) \geq c_2 q_n,$$

and

$$\mathbb{P}(X_i = 0 \mid X_{[n] \setminus \{i\}}) \geq c_2,$$

for some constant $c_2 = c_2(\beta) > 0$. Therefore,

$$\mathbb{P}(y_i \in V_{1,+}(G^*, X^n)) \geq c_2 q_n \mathbb{P}(y_i \in V_+(G^*, X^n)),$$

and

$$\mathbb{P}(y_i \in V_{0,+}(G^*, X^n)) \geq c_2 \mathbb{P}(y_i \in V_+(G^*, X^n)).$$

Since $q_n = \Theta(n^{\theta-1})$, we obtain

$$\mathbb{P}(y_i \in V_{1,+}(G^*, X^n)) \geq c_3 n^{\theta-1} n^{-(1-\epsilon/2)\theta} = c_3 n^{-1+\epsilon\theta/2},$$

and similarly

$$\mathbb{P}(y_i \in V_{0,+}(G^*, X^n)) \geq c_3 n^{-(1-\epsilon/2)\theta} \geq c_3 n^{-1+\epsilon\theta/2},$$

for all sufficiently large n .

The vertices $y_1, \dots, y_N$ have pairwise graph distance at least 5 in G^* . Therefore the events

$$\{y_i \in V_{0,+}(G^*, X^n)\}$$

depend on disjoint collections of Markov coordinates, except for possible nearest-neighbour dependencies along the line. By thinning the sequence $(y_i)_{i=1}^N$ into two parity classes and then retaining one class, we may extract a subset

$$I \subseteq [N], \quad |I| \geq N/2,$$

such that the corresponding local dependency neighborhoods are disjoint and separated by at least one coordinate. For this thinned family, the Markov property gives exponential concentration for the sum of the indicators

$$Z_0 := \sum_{i \in I} \mathbf{1}\{y_i \in V_{0,+}(G^*, X^n)\}.$$

Indeed,

$$\mathbb{E}[Z_0] \geq \frac{N}{2} c_3 n^{-1+\epsilon\theta/2} \geq c_4 n^{\epsilon\theta/2-\xi}.$$

By the choice $\xi < \epsilon\theta$, and after reducing ϵ by a constant factor if necessary,

$$\mathbb{E}[Z_0] \geq n^{c_5}$$

for some $c_5 > 0$. The Chernoff bound for bounded additive functionals of a finite-state Markov chain gives

$$\mathbb{P}(Z_0 \leq \ln n) \leq \exp\{-n^{c_6}\}$$

for some $c_6 > 0$.

The same argument applied to

$$Z_1 := \sum_{i \in I} \mathbf{1}\{y_i \in V_{1,+}(G^*, X^n)\}$$

gives

$$\mathbb{P}(Z_1 \leq \ln n) \leq \exp\{-n^{c_6}\}.$$

Therefore,

$$\mathbb{P}(|V_{0,+}(G^*, X^n)| \wedge |V_{1,+}(G^*, X^n)| \geq \ln n) = 1 - o(1).$$

Finally, since

$$V_{0,+}(G, X^n) = V_{0,+}(G^*, X^n), \quad V_{1,+}(G, X^n) = V_{1,+}(G^*, X^n)$$

with probability $1 - o(1)$, the proposition follows.

Lemma 4: [$N(\lambda)$ general bound] Let U^n and V^n be independent and identically distributed according to P_{U^n} . Define

$$J_n(\lambda) := -\frac{1}{n} \log \mathbb{P}(l(U^n, V^n) = \lambda k \mid (U^n, V^n) \in T_\epsilon \times T_\epsilon). \quad (14)$$

Then

$$\mathbb{E} \left[N_\epsilon(\lambda \mid U^n) \mid U^n \in T_\epsilon^{(n)} \right] \leq \frac{1}{1 - \delta_n} 2^{n(\tilde{H}_n + 3\epsilon - J_n(\lambda)) + o(n)}. \quad (15)$$

Let

$$\mathcal{H}(\lambda) := \left\{ (u^n, v^n) \in T_\epsilon^{(n)} \times T_\epsilon^{(n)} : l(u^n, v^n) = \lambda k \right\},$$

where $l(u^n, v^n)$ denotes the overlap count between u^n and v^n .

For each fixed $u^n \in T_\epsilon^{(n)}$, define the slice

$$\mathcal{H}(\lambda \mid u^n) := \left\{ v^n \in T_\epsilon^{(n)} : (u^n, v^n) \in \mathcal{H}(\lambda) \right\}.$$

We also note that the shorthand $N(\lambda)$ is a function of u^n , i.e.

$$N(\lambda) = N(\lambda \mid u^n) := |\mathcal{H}(\lambda \mid u^n)|.$$

Thus, $N(\lambda \mid u^n)$ is the number of sequences $v^n \in T_\epsilon^{(n)}$ whose overlap with the fixed sequence u^n is equal to λk .

The set $\mathcal{H}(\lambda)$ may be viewed as the collection of all admissible pairs (u^n, v^n) with normalized overlap λ . The slice $\mathcal{H}(\lambda | u^n)$ is then the collection of all neighbors of u^n inside this pair set. Summing the slice sizes over all $u^n \in T_\epsilon^{(n)}$ counts every pair in $\mathcal{H}(\lambda)$ exactly once. Therefore,

$$\sum_{u^n \in T_\epsilon^{(n)}} N(\lambda | u^n) = \sum_{u^n \in T_\epsilon^{(n)}} |\mathcal{H}(\lambda | u^n)| = |\mathcal{H}(\lambda)|.$$

We now use the AEP property of the typical set to justify replacing the conditional distribution of U^n given $U^n \in T_\epsilon^{(n)}$ by the uniform distribution on $T_\epsilon^{(n)}$.

For every $u^n \in T_\epsilon^{(n)}$, the definition of $T_\epsilon^{(n)}$ implies

$$2^{-n(\tilde{H}_n + \epsilon)} \leq P_{U^n}(u^n) \leq 2^{-n(\tilde{H}_n - \epsilon)}.$$

In this sense, the conditional distribution of U^n on $T_\epsilon^{(n)}$ is approximately uniform. To make this precise, define

$$p_T(u^n) := \mathbb{P}(U^n = u^n | U^n \in T_\epsilon^{(n)}), \quad u^n \in T_\epsilon^{(n)}.$$

For any $u^n \in T_\epsilon^{(n)}$,

$$p_T(u^n) = \frac{P_{U^n}(u^n)}{\mathbb{P}(U^n \in T_\epsilon^{(n)})}.$$

Hence,

$$p_T(u^n) \leq \frac{2^{-n(H - \epsilon)}}{\mathbb{P}(U^n \in T_\epsilon^{(n)})}.$$

Since

$$\mathbb{P}(U^n \in T_\epsilon^{(n)}) \geq 1 - \delta_n$$

with $\delta_n \rightarrow 0$, for every $u^n \in T_\epsilon^{(n)}$,

$$p_T(u^n) = \frac{P_{U^n}(u^n)}{\mathbb{P}(U^n \in T_\epsilon^{(n)})} \leq \frac{2^{-n(\tilde{H}_n - \epsilon)}}{1 - \delta_n}.$$

Hence,

$$\sum_{u^n \in T_\epsilon^{(n)}} p_T(u^n) N(\lambda | u^n) \leq \frac{2^{-n(\tilde{H}_n - \epsilon)}}{1 - \delta_n} |\mathcal{H}(\lambda)|.$$

Thus,

$$\mathbb{E}[N(\lambda | U^n) | U^n \in T_\epsilon^{(n)}] = \sum_{u^n \in T_\epsilon^{(n)}} p_T(u^n) N(\lambda | u^n) \tag{37}$$

$$\leq \frac{2^{-n(\tilde{H}_n - \epsilon)}}{1 - \delta_n} |\mathcal{H}(\lambda)|. \tag{38}$$

We now derive an upper bound on the quantity $|\mathcal{H}(\lambda)|$, where

$$\mathcal{H}(\lambda) := \left\{ (u^n, v^n) \in T_\epsilon^{(n)} \times T_\epsilon^{(n)} : l(u^n, v^n) = \lambda k \right\}.$$

To relate this counting quantity to a probability, let U^n and V^n be independent and identically distributed according to P_{U^n} , and define

$$r_n(\lambda) := \mathbb{P}(l(U^n, V^n) = \lambda k, U^n \in T_\epsilon^{(n)}, V^n \in T_\epsilon^{(n)}).$$

By expanding this probability over all pairs in $\mathcal{H}(\lambda)$, we obtain

$$r_n(\lambda) = \sum_{(u^n, v^n) \in \mathcal{H}(\lambda)} P_{U^n}(u^n) P_{U^n}(v^n). \tag{39}$$

Now fix any pair $(u^n, v^n) \in \mathcal{H}(\lambda)$. Since both u^n and v^n belong to $T_\epsilon^{(n)}$, the definition of the typical set implies

$$P_{U^n}(u^n) \leq 2^{-n(\tilde{H} - \epsilon)}, \quad P_{U^n}(v^n) \leq 2^{-n(\tilde{H} - \epsilon)},$$

where $\tilde{H} := \frac{1}{n} H(U^n)$. Hence

$$P_{U^n}(u^n) P_{U^n}(v^n) \leq 2^{-2n(\tilde{H}-\epsilon)}$$

for every $(u^n, v^n) \in \mathcal{H}(\lambda)$.

Similarly, for every $(u^n, v^n) \in \mathcal{H}(\lambda)$,

$$P_{U^n}(u^n) P_{U^n}(v^n) \geq 2^{-2n(\tilde{H}+\epsilon)}.$$

Therefore,

$$r_n(\lambda) \geq |\mathcal{H}(\lambda)| 2^{-2n(\tilde{H}+\epsilon)}. \quad (40)$$

Rearranging gives the desired upper bound

$$|\mathcal{H}(\lambda)| \leq 2^{2n(\tilde{H}+\epsilon)} r_n(\lambda). \quad (41)$$

Thus, the number of typical pairs having overlap λk is controlled by the probability that two independent draws from the source are both typical and have overlap λk .

We now substitute this bound into the estimate for the conditional expectation of $N(\lambda | U^n)$. Recall that

$$\mathbb{E} \left[N(\lambda | U^n) | U^n \in T_\epsilon^{(n)} \right] = \sum_{u^n \in T_\epsilon^{(n)}} p_T(u^n) N(\lambda | u^n) \quad (42)$$

$$\leq \frac{2^{-n(\tilde{H}-\epsilon)}}{1 - \delta_n} |\mathcal{H}(\lambda)|, \quad (43)$$

where

$$p_T(u^n) = \mathbb{P}(U^n = u^n | U^n \in T_\epsilon^{(n)})$$

and $\mathbb{P}(U^n \in T_\epsilon^{(n)}) \geq 1 - \delta_n$.

Using the bound on $|\mathcal{H}(\lambda)|$, we obtain

$$\mathbb{E} \left[N(\lambda | U^n) | U^n \in T_\epsilon^{(n)} \right] \leq \frac{2^{-n(\tilde{H}-\epsilon)}}{1 - \delta_n} \cdot 2^{2n(\tilde{H}+\epsilon)} r_n(\lambda) \quad (44)$$

$$= \frac{2^{n(\tilde{H}+3\epsilon)}}{1 - \delta_n} r_n(\lambda). \quad (45)$$

Hence,

$$\mathbb{E} \left[N(\lambda | U^n) | U^n \in T_\epsilon^{(n)} \right] \leq \frac{2^{n(\tilde{H}+3\epsilon)}}{1 - \delta_n} r_n(\lambda). \quad (46)$$

We now express $r_n(\lambda)$ in terms of the overlap probability under the conditional law on the typical set. Recall that

$$r_n(\lambda) := \mathbb{P}(l(U^n, V^n) = \lambda k, U^n \in T_\epsilon^{(n)}, V^n \in T_\epsilon^{(n)}).$$

Since U^n and V^n are independent,

$$\mathbb{P}(U^n \in T_\epsilon^{(n)}, V^n \in T_\epsilon^{(n)}) = \mathbb{P}(U^n \in T_\epsilon^{(n)})^2.$$

Therefore,

$$\begin{aligned} r_n(\lambda) &= \mathbb{P}(U^n \in T_\epsilon^{(n)}, V^n \in T_\epsilon^{(n)}) \mathbb{P}(l(U^n, V^n) = \lambda k | U^n, V^n \in T_\epsilon^{(n)}) \\ &= \mathbb{P}(U^n \in T_\epsilon^{(n)})^2 \mathbb{P}(l(U^n, V^n) = \lambda k | U^n, V^n \in T_\epsilon^{(n)}). \end{aligned} \quad (47)$$

Since

$$\mathbb{P}(U^n \in T_\epsilon^{(n)}) \leq 1,$$

we have the upper bound

$$r_n(\lambda) \leq \mathbb{P}(l(U^n, V^n) = \lambda k | U^n, V^n \in T_\epsilon^{(n)}).$$

Hence

$$r_n(\lambda) \leq 2^{-nJ_n(\lambda)}.$$

Substituting this bound into the preceding estimate gives

$$\begin{aligned}
\mathbb{E} \left[N(\lambda \mid U^n) \mid U^n \in T_\epsilon^{(n)} \right] &\leq \frac{2^{n(\tilde{H}_n + 3\epsilon)}}{1 - \delta_n} r_n(\lambda) \\
&\leq \frac{2^{n(\tilde{H}_n + 3\epsilon)}}{1 - \delta_n} 2^{-nJ_n(\lambda)} \\
&= \frac{1}{1 - \delta_n} 2^{n(\tilde{H}_n + 3\epsilon - J_n(\lambda))}.
\end{aligned} \tag{48}$$

Therefore,

$$\mathbb{E} \left[N(\lambda \mid U^n) \mid U^n \in T_\epsilon^{(n)} \right] \leq \frac{1}{1 - \delta_n} 2^{n(\tilde{H}_n + 3\epsilon - J_n(\lambda))}. \tag{49}$$